

Digital Signal Processing - Lecture 3

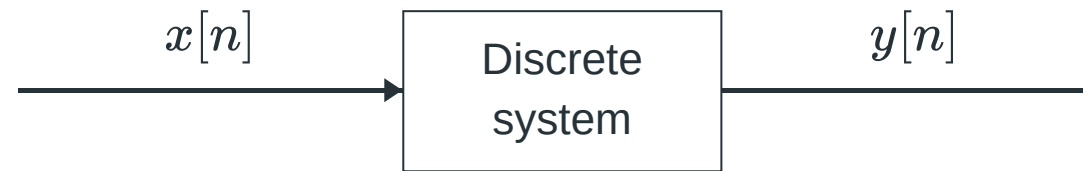
Discrete Fourier transform

Ewout Bergsma

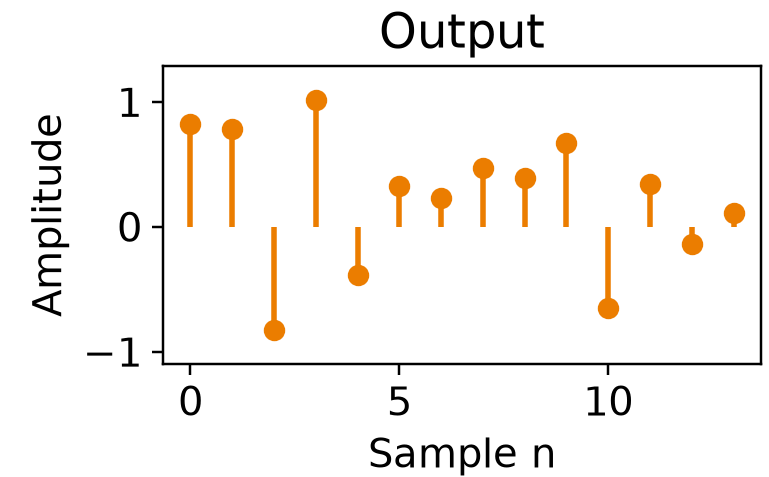
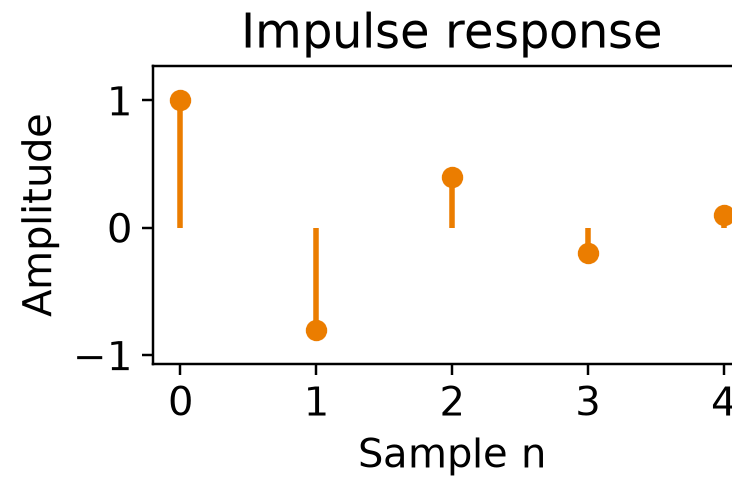
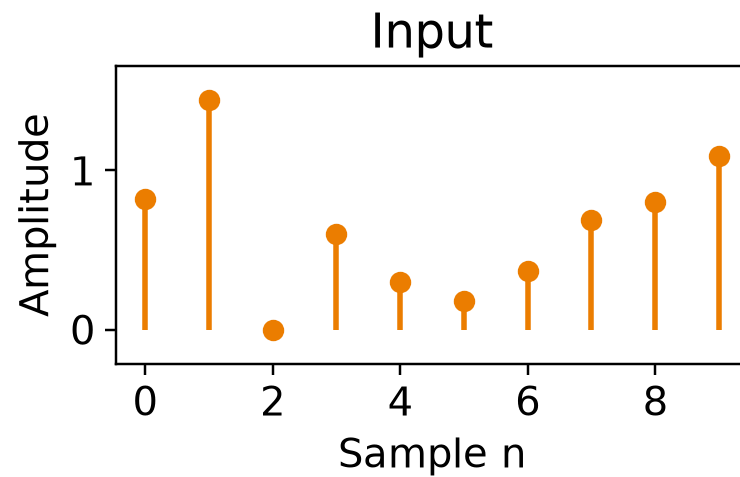
Hanze University of Applied Sciences

Overview

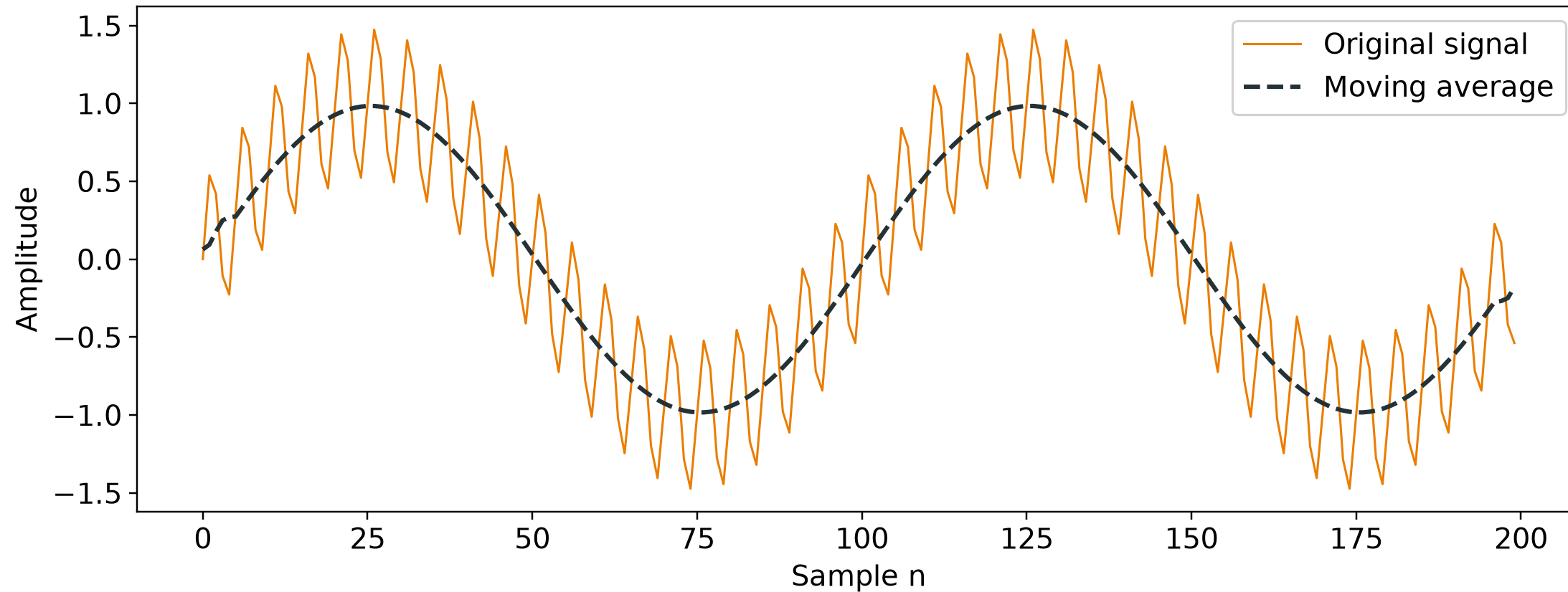
Last time: input, system, output



$$x[n] * h[n] = y[n]$$

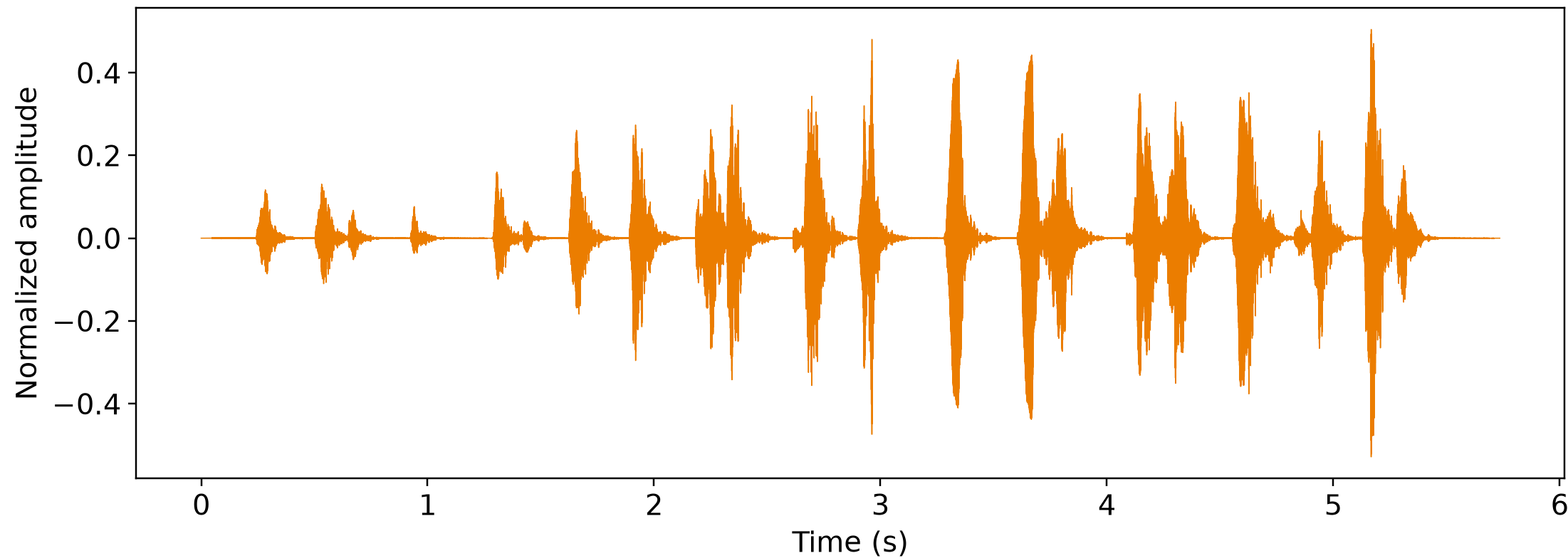


Last time: moving average



You did your exercises, right?

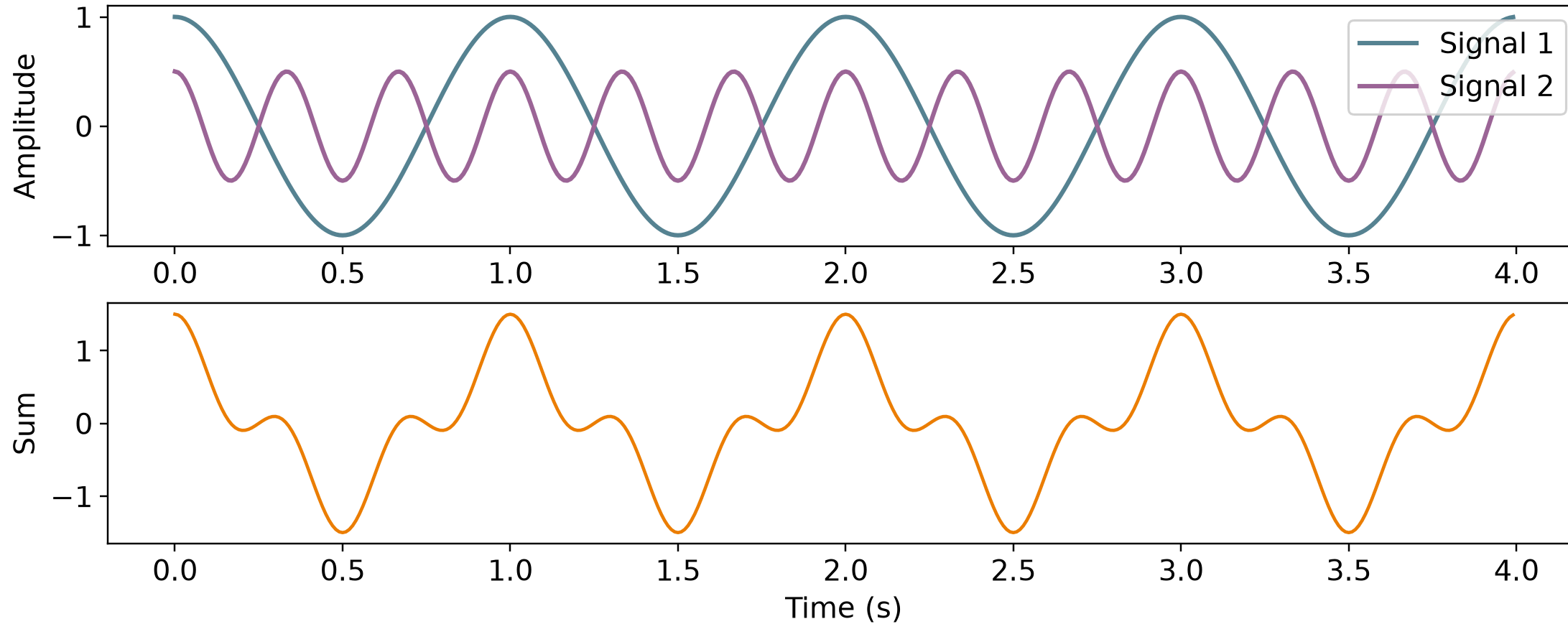
A real-world signal



$y[n]$ is observed. What might $x[n]$ and $h[n]$ be?

Source: <https://bigsoundbank.com/detail-2776-cockatiel-parakeet-8.html>

From a sum to its components



Can we identify the frequencies present in the combined signal?

This lecture

01 Frequency domain

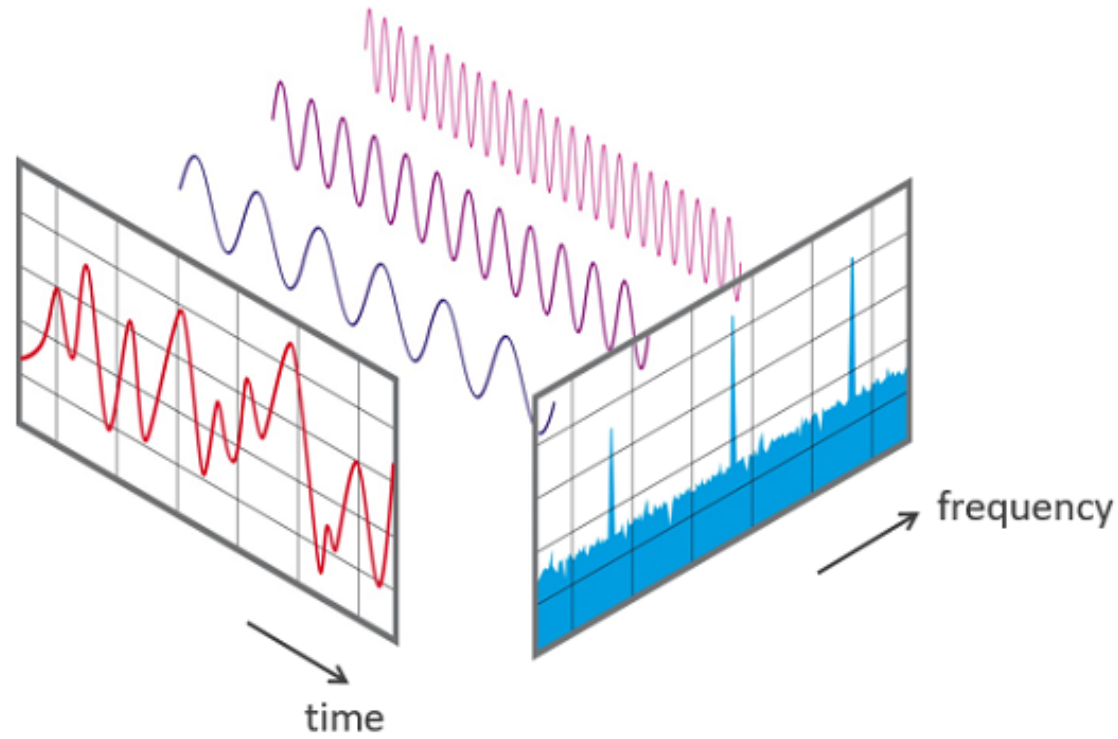
02 Discrete Fourier transform

03 Understanding the equation

04 Exercises

| Frequency domain

Time domain and frequency domain



Source: <https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft>

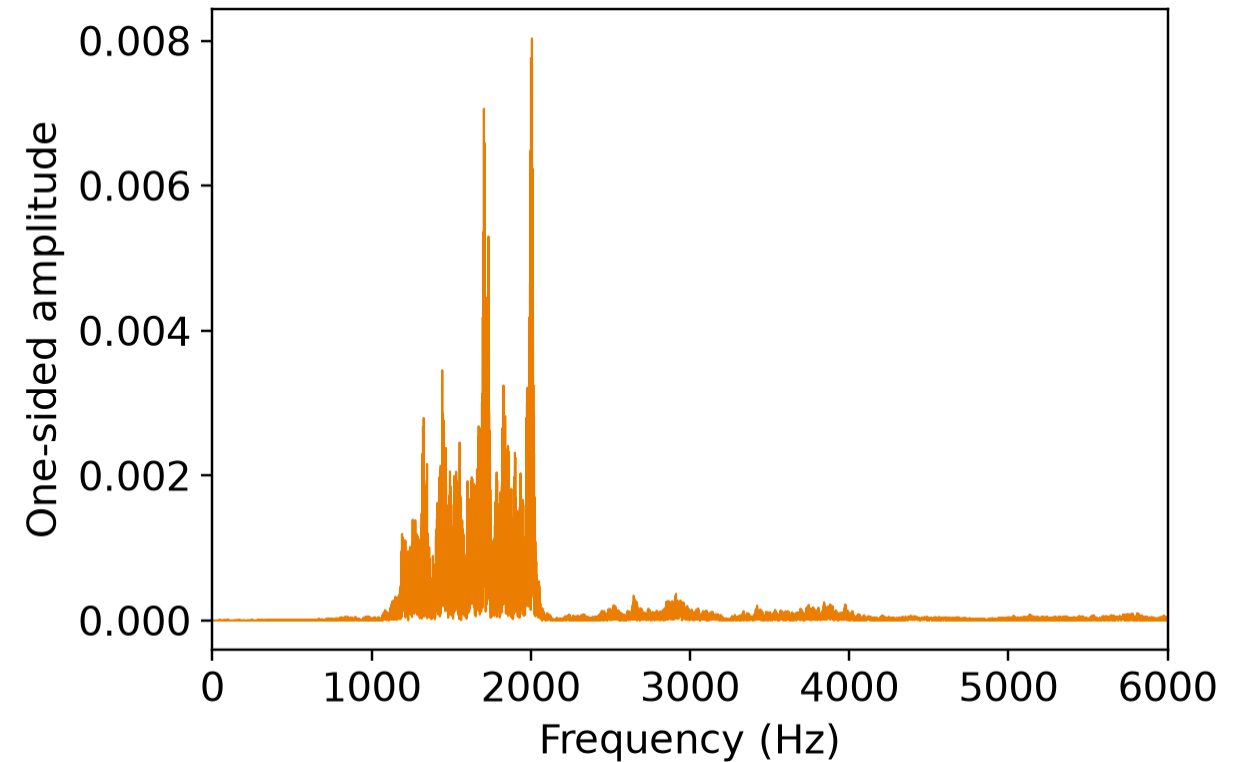
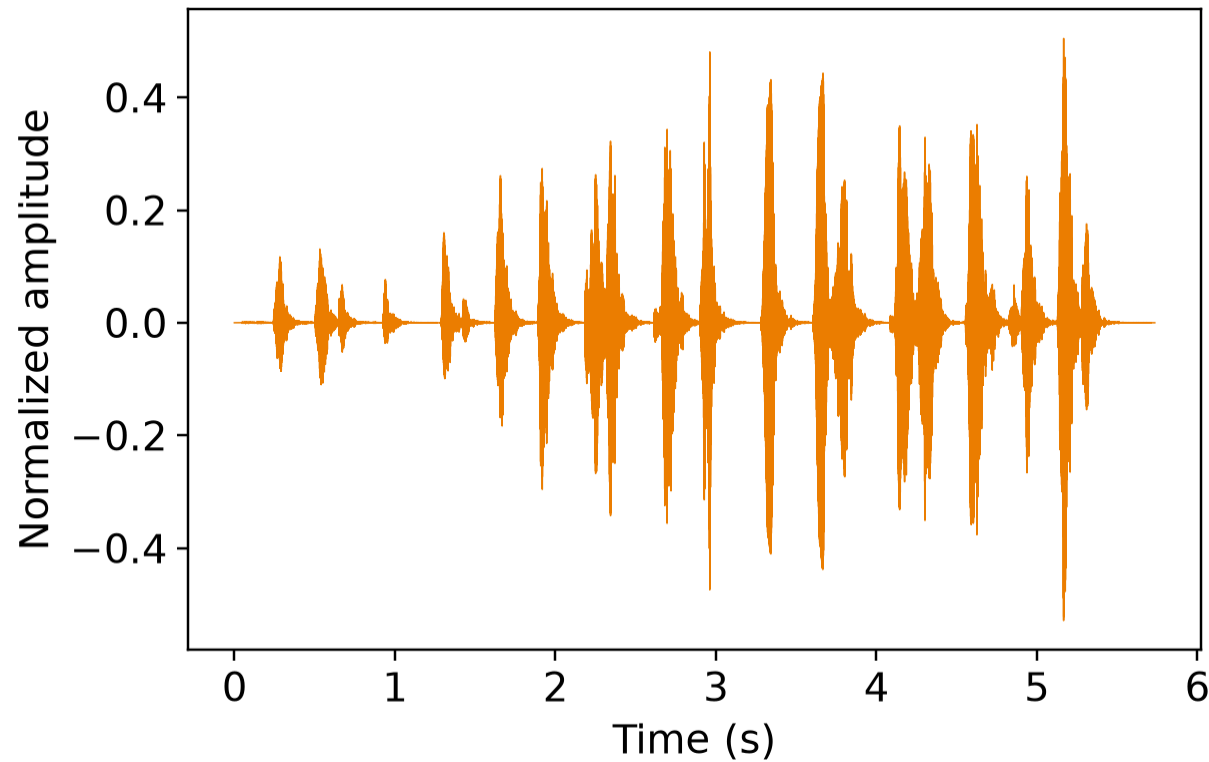
Jean-Baptiste Joseph Fourier



Jean-Baptiste Joseph Fourier

Source: https://en.wikipedia.org/wiki/Joseph_Fourier

An audio signal and its spectrum



Source: <https://bigsoundbank.com/detail-2776-cockatiel-parakeet-8.html>

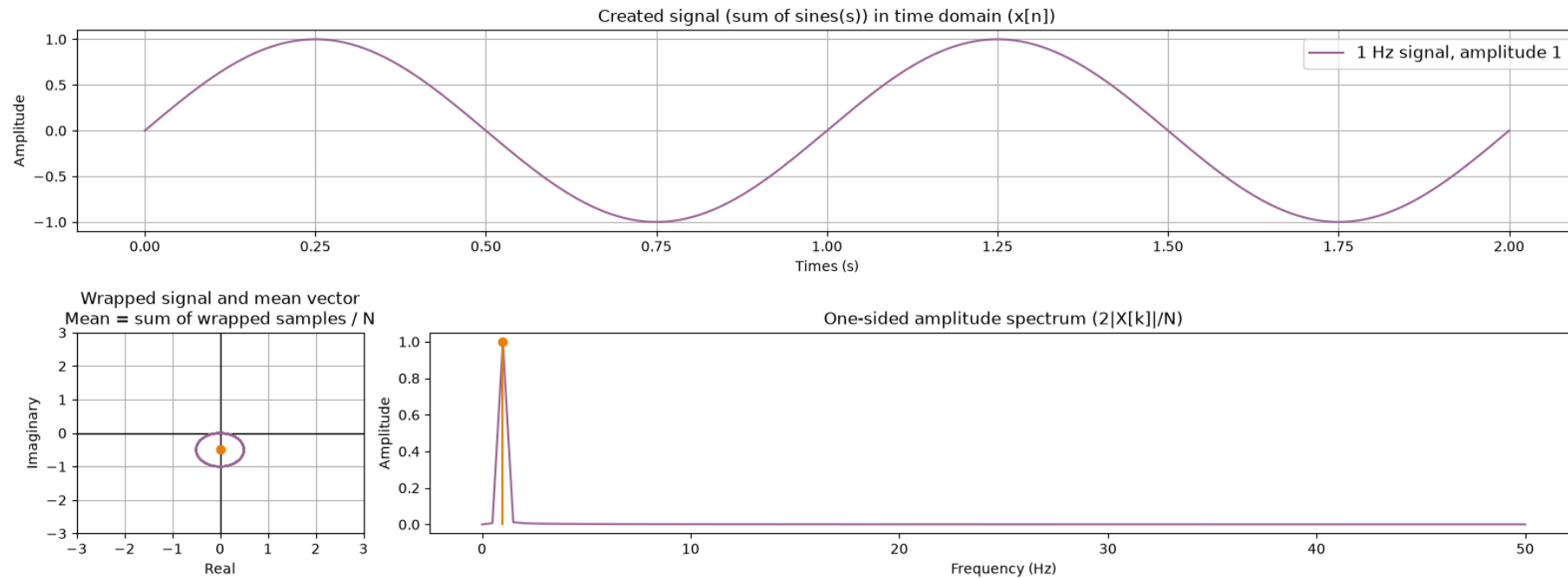
| Discrete Fourier transform

Discrete Fourier transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

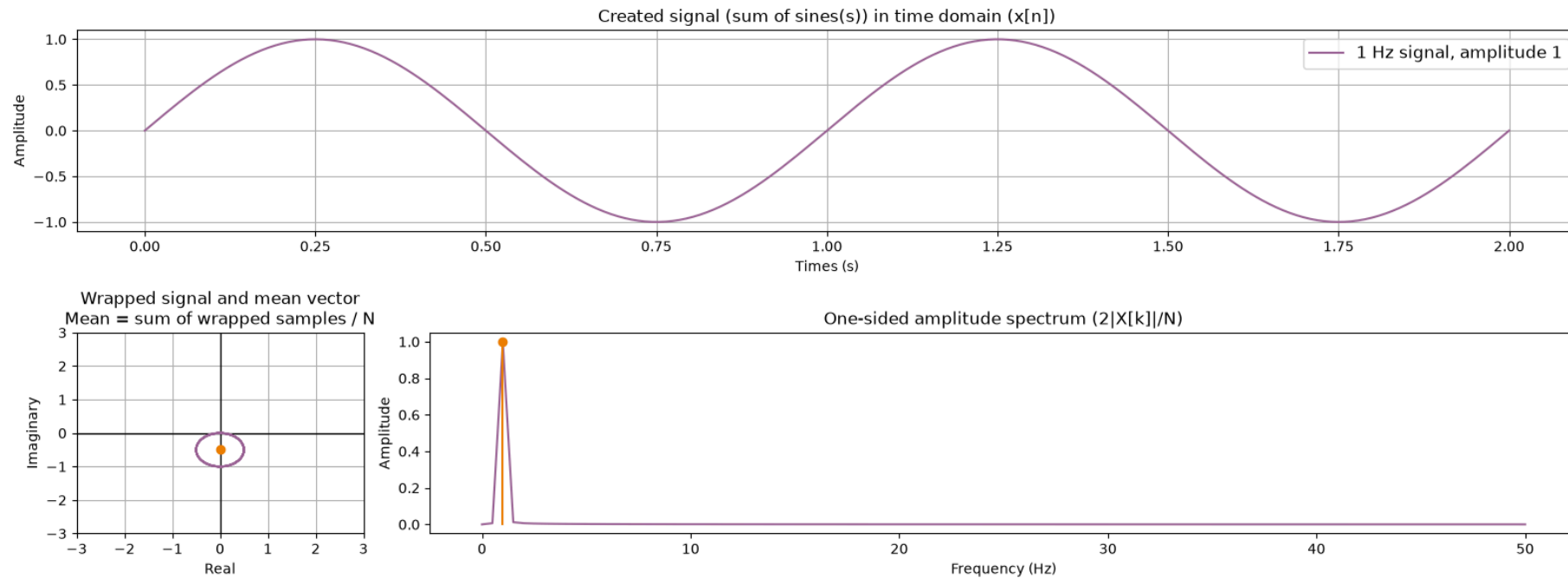
- $x[n]$ is a sequence of N samples.
- $X[k]$ contains complex frequency-domain coefficients.

Fourier visualisation



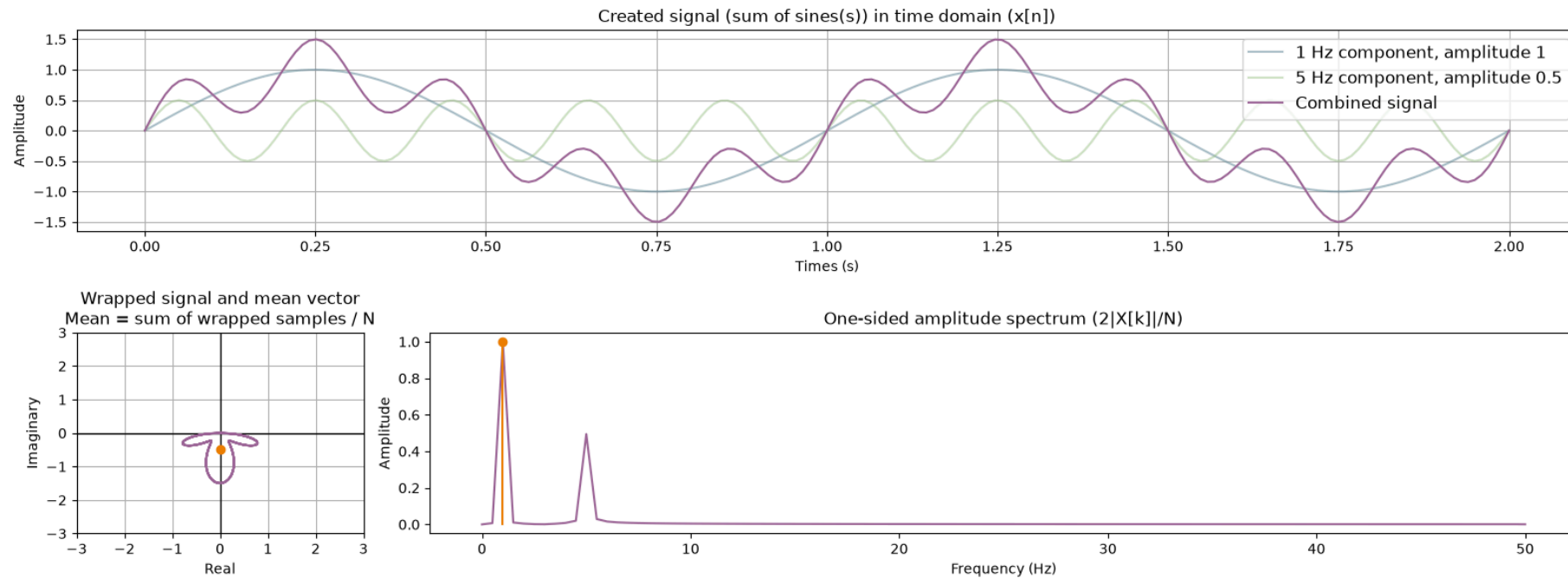
Source: <https://github.com/EwoutBergsma/Interactive-Fourier-Transform-Visualisation>

Matching a frequency



Source: <https://github.com/EwoutBergsma/Interactive-Fourier-Transform-Visualisation>

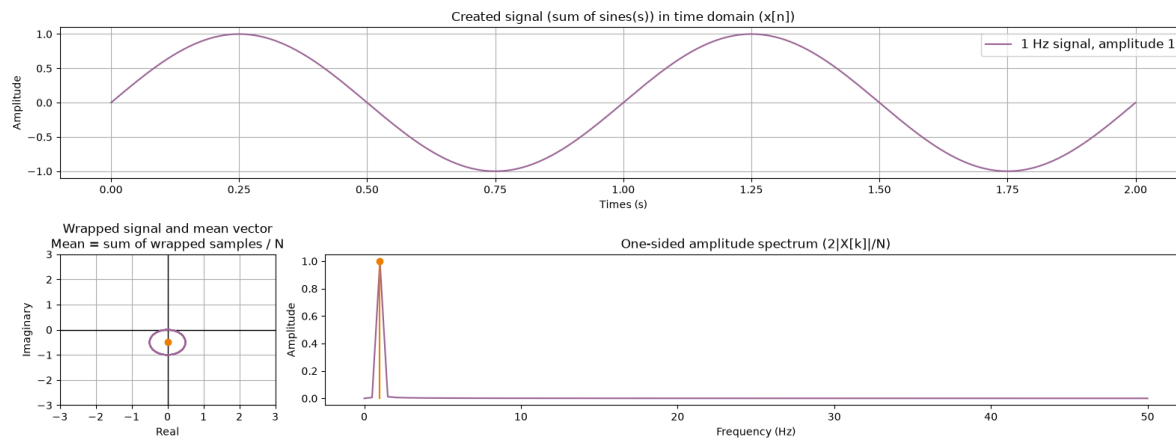
Separating two frequencies



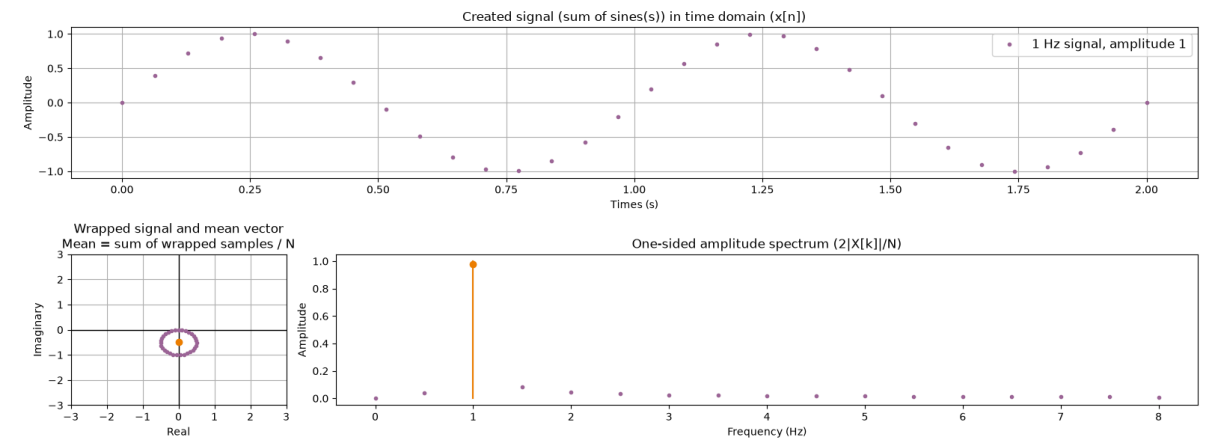
Source: <https://github.com/EwoutBergsma/Interactive-Fourier-Transform-Visualisation>

From wrapped samples to DFT bins

Continuous curves



Discrete data points



Source: <https://github.com/EwoutBergsma/Interactive-Fourier-Transform-Visualisation>

| Understanding the equation

Time index and frequency index

$$\underbrace{X[k]}_{\text{frequency-domain}} = \sum_{n=0}^{N-1} \underbrace{x[n]}_{\text{time-domain}} e^{-j2\pi kn/N}$$

- n indexes time samples.
- k indexes DFT bins.

The input samples

$$X[k] = \sum_{n=0}^{N-1} \underbrace{x[n]}_{\substack{\text{input} \\ \text{sample}}} e^{-j2\pi kn/N}$$

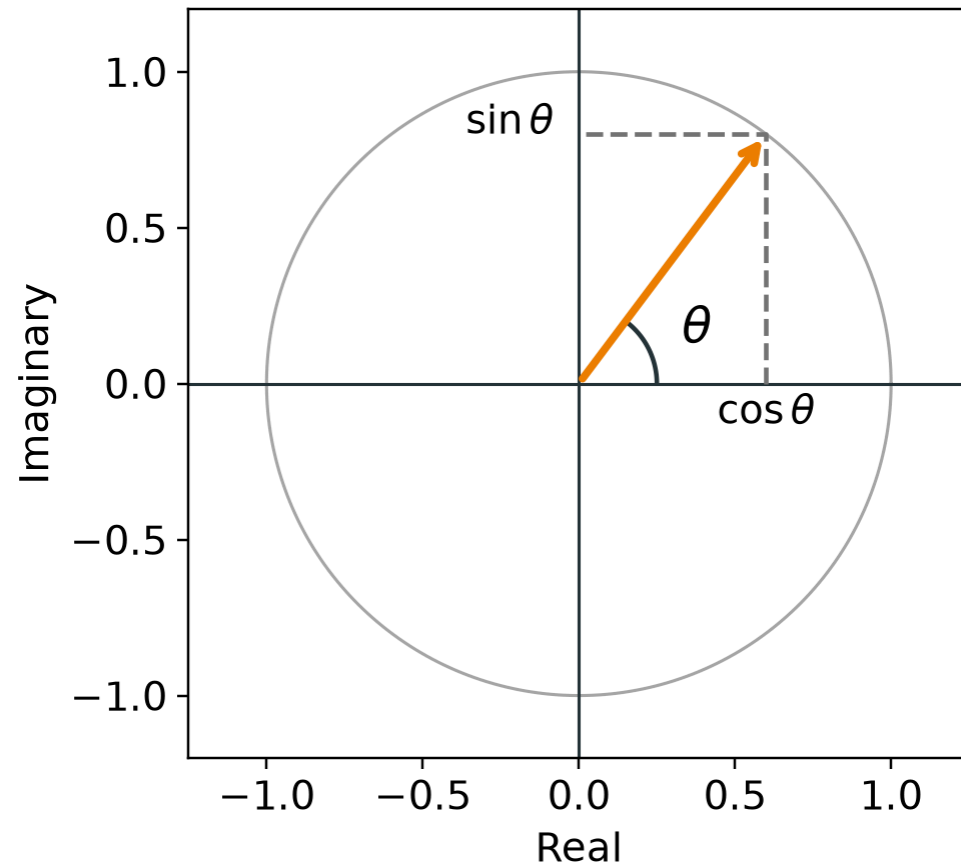
Each sample contributes to every frequency coefficient.

The complex exponential

$$X[k] = \sum_{n=0}^{N-1} x[n] \underbrace{e^{-j2\pi kn/N}}_{\text{complex sinusoid}}$$

Review complex numbers and Euler's formula if needed.

Euler's formula



<https://www.khanacademy.org/math/algebra-home/alg-complex-numbers>

One full rotation

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \underbrace{2\pi}_{\text{one rotation}} kn/N}$$

An angle of 2π radians corresponds to one revolution of the unit circle.

The negative sign

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

The forward DFT uses the negative-exponent convention.

Position within the sample window

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi k \underbrace{n/N}_{\text{fraction of window}}}$$

n/N indicates the sample position as a fraction of the N -sample window.

Selecting a frequency bin

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

- k selects the frequency bin and controls the wrapping frequency.
- Increasing k increases the phase change between consecutive samples.

Sum over the samples

$$X[k] = \sum_{\substack{n=0 \\ \underbrace{\hspace{1cm}} \\ \text{all } N \text{ samples}}}^{N-1} x[n] e^{-j2\pi kn/N}$$

For each k , multiply the samples by that complex sinusoid and sum the results.

Questions?

| Exercises

Exercises!

Explore your signal in the frequency domain.

`../exercises/DSP-Lecture-03-Exercises.ipynb`